Tests for divisibility

A Cambridge insert: deciding what divides a number without doing the division.

LESSON 39 | 1 × 40 MIN | MATEMATIKA 6, QO'SHIMCHA MAVZU | S7 1.5

LESSON CLOCK

12' | 10' | 12' | 6'

The tests	12 min
Using them together	10 min
Factorising quickly	12 min
Why the test for 3 works	6 min

LEARNING OBJECTIVES

- State and apply the tests for 2, 3, 4, 5, 6, 8, 9 and 10.
- Combine tests to check divisibility by a composite number.
- Use the tests to factorise quickly.
- Explain why the test for 3 works.

TEXTBOOK

National	pp. 103–105
Cambridge	Stage 7, pp. 24–27
Workbook	Exercise 1.5

01 Explanation

The tests

DIVISOR	TEST	EXAMPLE
2	last digit even	374
3	digit sum divisible by 3	$471 \to 12$
4	last two digits divisible by 4	$1316 \to 16$
5	last digit 0 or 5	385
6	divisible by 2 and by 3	474
8	last three digits divisible by 8	$3128 \to 128$
9	digit sum divisible by 9	$576 \to 18$
10	last digit 0	4370

THE DIGIT-SUM TESTS CAN BE REPEATED

For $99\,999$ the digit sum is 45, whose digit sum is 9 — divisible by 9, so the original is too.

Using them together

For a composite divisor, test its coprime factors separately.

DIVISOR	TEST	WARNING
6	2 and 3	—
12	3 and 4	not 2 and 6
15	3 and 5	—
18	2 and 9	not 3 and 6

THE TWO FACTORS MUST SHARE NOTHING

$12 = 2 \cdot 6$, but passing the 2 and 6 tests is not enough: 18 passes both and is not a multiple of 12. Use 3 and 4, which are coprime.

Factorising quickly

NUMBER	TESTS IT PASSES	FACTORISATION
180	$2, 3, 4, 5, 9, 10$	$2^2 \cdot 3^2 \cdot 5$
252	$2, 3, 4, 9$	$2^2 \cdot 3^2 \cdot 7$
385	5	$5 \cdot 7 \cdot 11$
1001	none of the small ones	$7 \cdot 11 \cdot 13$

THE TESTS SAVE THE TRIAL DIVISIONS

Knowing at a glance that 180 is divisible by 4 and by 9 gives $180 = 36 \cdot 5$ immediately, without any long division at all.

Why the test for 3 works

Every power of ten is one more than a multiple of 3: $10 = 9 + 1$, $100 = 99 + 1$, $1000 = 999 + 1$.

STEP	FOR 471
split by place value	$4 \cdot 100 + 7 \cdot 10 + 1$
rewrite the powers	$4(99 + 1) + 7(9 + 1) + 1$
separate	$(4 \cdot 99 + 7 \cdot 9) + (4 + 7 + 1)$
the first bracket	always a multiple of 3
so the test	depends only on the digit sum

THE SAME ARGUMENT WORKS FOR 9

99 and 999 are multiples of 9 as well, which is why the digit-sum test serves both numbers — and why no such simple test exists for 7.

02 Worked examples

EXAMPLE 1 **Is 471 divisible by 3? By 9?**

1. Digit sum: $4 + 7 + 1 = 12$.
2. 12 is divisible by 3 — so 471 is.
3. 12 is not divisible by 9 — so 471 is not.

ANSWER

By 3 yes, by 9 no

EXAMPLE 2 **Is *1316* divisible by *4*?**

1. Look at the last two digits: 16.
2. $16 \div 4 = 4$.
3. So yes.

ANSWER

Yes

EXAMPLE 3 **Factorise *180* using the tests.**

1. Last two digits 80 — divisible by 4.
   $180 = 4 \cdot 45$
2. Digit sum 9 — divisible by 9.
   $45 = 9 \cdot 5$
3. $180 = 4 \cdot 9 \cdot 5$
4. $= 2^2 \cdot 3^2 \cdot 5$

ANSWER

$2^2 \cdot 3^2 \cdot 5$

03 Interactive model

Call out five-digit numbers and have the class shout which of 2, 3, 5 and 9 divide them; the tests become automatic in a single lesson.

Which numbers divide it?

Is 374 divisible by 2?

- yes
- no

Is 471 divisible by 3?

- yes
- no

Is 471 divisible by 9?

- yes
- no

Is 1316 divisible by 4?

- yes
- no

Is 385 divisible by 5?

- yes
- no

Is *474* divisible by *6*?

- yes
- no

Is *576* divisible by *9*?

- yes
- no

To test for *12* use:

- *2* and *6*
- *3* and *4*
- *12* itself only
- *6* and *6*

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Divisible	Bo‘linadi	Делится
Test for divisibility	Bo‘linish alomati	Признак делимости
Digit sum	Raqamlar yig‘indisi	Сумма цифр
Last digit	Oxirgi raqam	Последняя цифра
Even number	Juft son	Чётное число
Odd number	Toq son	Нечётное число
Composite number	Murakkab son	Составное число
Remainder	Qoldiq	Остаток

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A number is divisible by 3 when:

- it is even
- its digit sum is
- it ends in 3
- it is odd

By 4 when:

- it is even
- its last two digits are
- its digit sum is
- it ends in 4

By 6 when it is divisible by:

- 2 and 3
- 3 only
- 2 only
- 6 alone

By 9 when:

- it ends in 9
- its digit sum is
- it is odd
- it is even

Testing for 12 by 2 and 6 is:

- correct
- not enough
- too strong
- impossible

Which has no simple digit test?

- 3
- 7
- 9
- 5

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Is *374* divisible by *2*?

   ANSWER Yes

2. Is *471* divisible by *3*?

   ANSWER Yes

3. Is *471* divisible by *9*?

   ANSWER No

4. Is *1316* divisible by *4*?

   ANSWER Yes

5. Is *385* divisible by *5*?

   ANSWER Yes

6. Is *4370* divisible by *10*?

   ANSWER Yes

7. Is *576* divisible by *9*?

   ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. Is 474 divisible by 6?

   ANSWER Yes

2. Is 3128 divisible by 8?

   ANSWER Yes

3. Which of $2, 3, 5, 9$ divide 180?

   ANSWER All four

4. Which divide 385?

   ANSWER 5 only

5. Factorise 180

   ANSWER $2^2 \cdot 3^2 \cdot 5$

6. Factorise 252

   ANSWER $2^2 \cdot 3^2 \cdot 7$

7. Test for 15: use

   ANSWER 3 and 5

Hard · stretch and reason

7 PROBLEMS

1. Is $99\,999$ divisible by 9?

   ANSWER Yes — digit sum 45

2. Why is testing 12 with 2 and 6 wrong?

   ANSWER 18 passes both but is not a multiple of 12

3. Factorise 1001

   ANSWER $7 \cdot 11 \cdot 13$

4. Explain the test for 3

   ANSWER Every power of ten is one more than a multiple of 3

5. The smallest digit d making $47d$ divisible by 9

   ANSWER $d = 7$

6. Which digits d make $12d$ divisible by 4?

   ANSWER $0, 4, 8$

7. Is $2 \cdot 3 \cdot 5 \cdot 7 + 1$ divisible by any of $2, 3, 5, 7$?

   ANSWER No — each leaves remainder 1

07 Homework

Homework — 5 tasks

Write which test you used beside every answer.

1. Test $2\,358$ for divisibility by $2, 3, 4, 6$ and 9.
2. Test $7\,425$ for divisibility by $3, 5, 9$ and 15.
3. Factorise 360 using the tests.
4. Find the digit d making $61d$ divisible by 3, listing every possibility.
5. Explain in two sentences why the digit-sum test works for 9.